Academy of Interdisciplinary Studies,
The Hong Kong University of
Science and Technology

Mingpei CANG

PhD Candidate

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Overview

Background in ***Solid Mechanics*** with a focus on 3D printing and mechanical metamaterials. Experienced in ***systematic, detail-oriented*** engineering workflows, including ***structural design***, finite element analysis (***FEA***), additive manufacturing (AM) ***process optimization***, and ***mechanical testing***. Familiar with vat photopolymerization AM technologies and hands-on experience in designing and building 3D printing systems.

Education

The Hong Kong University of Science and Technology (HKUST) Sep 2021 – Aug 2026
PhD in Smart Manufacturing Co-supervised by Dr. Huachen Cui and Prof. Michael Yu Wang

Dalian University of Technology Sep 2017 – Jun 2021
BEng in Engineering Mechanics GPA: 90.1/100, Ranking: 3/33 (Qian Lingxi Elite Program)

Selected Projects

Design of High-Stiffness, High-Strength Perforated Metallic Plate Lattice Structures Dec 2021 – Dec 2023

- Developed a **Rayleigh quotient–based design** criterion to optimize perforation placement in low-density plate lattice structures for maximum buckling strength.
- Developed **parametric designs** of SC, BCC, and FCC perforated plate lattice structures and performed **nonlinear FEA** in Abaqus, achieving a 15.1% increase in buckling strength.
- Collaborated cross-functionally on μ-LPBF fabrication and **quasi-static compression testing**, validating improved stiffness and strength of the designed structures.

Buckling strength optimization of variable-section truss lattice structures Dec 2020 – Sep 2022

- Developed variable-section strut designs for low-density truss lattice structures using **implicit modeling** and shape optimization, achieving a >25% increase in **buckling strength**.
- Built a **fully automated** Python-Abaqus workflow for modeling, simulation, and optimization, generating optimal designs across varying boundary conditions.
- Experimentally validated optimized designs via SLS nylon 3D printing and **quasi-static compression testing**, demonstrating strong agreement with FEA predictions.

Polarization imaging-based in-situ process monitoring for stereolithography 3D printing Sep 2023 – Present

- Designed a **compact polarization imaging module** for in-situ quality monitoring, integrated into 3D printing systems.
- Developed an image processing algorithm for real-time monitoring without compromising printing speed.
- Achieved **high-resolution defect detection** including dimensional deviations and internal voids, outperforming existing in-situ monitoring systems for stereolithography 3D printing.

Process optimization of high-resolution projection stereolithography 3D printing Apr 2024 – Aug 2025

- Developed a **numerical model** of the scattering–reaction–diffusion process in projection stereolithography to guide process optimization.
- Proposed a **hardware-agnostic printing process** compatible with materials from photopolymers to ceramics and scalable across systems with 2–50 μm optical resolution.
- Enabled integrated, pre-assembled fabrication of microscale transmission mechanisms with assembly clearances exceeding current additive manufacturing limits.

Curvature analysis and structural design of bone-mimicking lattice scaffolds Apr 2024 – Sep 2024

- Analyzed μ-CT bone scans to extract curvature and interfacial **geometry features** for scaffold design using Libigl.
- Designed TPMS-based lattice structures using nTop **parametric modeling** to match bone-like curvature characteristics and porosity.
- Collaborated cross-functionally with partner research groups on sample fabrication and in vivo testing, demonstrating improved osteogenesis compared to controls.

Conferences & Publications

- Solid Freeform Fabrication 2025, Austin, TX, United States — Oral presentation
- Material Research Society Fall Meeting 2024, Boston, MA, United States — Poster presentation
- Achieving sub-pixel level feature size and continuous printing for vat photopolymerization, **M. Cang** et al., under review at *Int. J. Extrem. Manuf.* — Journal Article
- Biocompatible piezoelectric lattice materials with ultrasound-regulated multimodal responses, A. Chen#, J. Su#, M. Zhou#, **M. Cang** et al., *Materials Science and Engineering: R: Reports*, (2025) 100876. — Journal Article

- Metallic perforated plate lattices with superior buckling strength, L. Zhang[#], **M. Cang**[#], J. Ding et al., *Materials and Design*, (2025) 113544. (***Featured on cover image***) Journal Article
- An efficient method for design of lattice core sandwich structures with superior buckling strength, **M. Cang**[#], L. Zhang[#], Y. Wang et al., *Engineering Optimization*, (2023) 1-19. Journal Article

Teaching & Technical Experience

Project Mentor - Undergraduate Research Project Advisor: Dr. Huachen Cui

- Dual Resolution Projection Stereolithography System Jul 2025 – Present
  Led a team of two undergraduate students to design and build a multi-resolution projection stereolithography system from scratch, integrating hardware and software development.

Teaching Assistant - Graduate Courses

- **Additive Manufacturing Fundamentals** (Instructor: Dr. Huachen Cui) Sep 2023 – Jun 2024
  Guided students in assembling FDM printers and demonstrated SLA 3D printing principles and operation.
- **Topology Optimization and Additive Manufacturing** (Instructor: Prof. Michael Yu Wang) Sep 2021 – Jan 2022
  Assisted students with finite element analysis (FEA) and topology optimization in Ansys.

Peer Reviewer for Journal Articles

- Materials Science and Engineering: R: Reports Dec 2024
- Thin-Walled Structures Mar 2025

Leadership & Extracurricular Experience

Animal Rescue Student Club *HKUST (Guangzhou)* Sep 2023 – Jul 2025
Founder & President

- Implemented humane and data-informed strategies to manage the campus stray cat population, including neutering, adoption, and community feeding programs.

Graduates and Alumni Association *HKUST* Sep 2021 – May 2022
Assistant Director, New Media Office

- Organized student activities and coordinated the distribution of anti-pandemic supplies during COVID-19.

Student Union *Dalian University of Technology* Jul 2019 – Jun 2020
Director, Study Department, Faculty of Vehicle Engineering and Mechanics (FVEM)

- Organized mechanics competitions, including theoretical mechanics and mechanics of materials.
- Contributed to the annual Undergraduate Teaching Quality Report for FVEM.

Awards

HKUST postgraduate Scholarship Sep 2021 – Aug 2025
Outstanding graduates award of Dalian city Jun 2021
China national scholarship Oct 2019

Skills

Structural Design & Simulation:
Parametric design, Implicit modeling, Finite element analysis (FEA), Nonlinear FEA, Buckling analysis, Abaqus, COMSOL

CAD & Computational Design software:
nTop, Autodesk Inventor, Fusion 360, FreeCAD

Programming & Automation:
Python, MATLAB,
Automated simulation and optimization workflows

Additive Manufacturing & Mechanics:
Vat photopolymerization, Selective laser sintering, Process optimization, Mechanical testing, Statistical analysis

LinkedIn

Personal Website
Project details

ORCID
Publication list